Lesson Planning

Complete Lesson Plan Book

Class: Class 12 – FSc Pre-Engineering

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Functions and Limits

Learning Objectives

• Define and explain domain and range in the context of Functions and Limits.

• Differentiate between types of functions and limit concept with suitable examples.

• Solve numerical problems related to theorems on limits accurately.

• Apply the ideas of Functions and Limits to everyday situations involving continuity.

Key Concepts

• Meaning, definition and scope of domain and range.

• Important properties, types and examples of types of functions.

• Relationship between limit concept and theorems on limits.

• Practical applications of Functions and Limits, especially continuity.

• Key terms and a summary of the main points of Functions and Limits.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Functions and Limits with a daily-life example.
2. Explain domain and range on the board with definitions and clear diagrams.
3. Discuss types of functions and limit concept using simple, relatable examples.
4. Involve the students through short questions on theorems on limits.
5. Solve a numerical on continuity together with the class.
6. Summarize the lesson and highlight the key points of Functions and Limits.

Classroom Activities

• Group discussion: students compare types of functions with limit concept.

• Board activity: solve a question on theorems on limits in pairs.

• Quick quiz on the key terms of Functions and Limits.

Assessment Questions

1. Define domain and range.
2. Differentiate between types of functions and limit concept.
3. Solve a numerical based on theorems on limits.
4. Write a short note on continuity.

Homework

• Read Functions and Limits from the book and solve the exercise questions.

• Prepare short notes on domain and range and types of functions.

Time Allocation

4–6 periods of 40 minutes each

Unit 2: Differentiation

Learning Objectives

• Define and explain derivative definition in the context of Differentiation.

• Differentiate between rules of differentiation and chain rule with suitable examples.

• Solve numerical problems related to implicit differentiation accurately.

• Apply the ideas of Differentiation to everyday situations involving higher order derivatives.

Key Concepts

• Meaning, definition and scope of derivative definition.

• Important properties, types and examples of rules of differentiation.

• Relationship between chain rule and implicit differentiation.

• Practical applications of Differentiation, especially higher order derivatives.

• Key terms and a summary of the main points of Differentiation.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Differentiation with a daily-life example.
2. Explain derivative definition on the board with definitions and clear diagrams.
3. Discuss rules of differentiation and chain rule using simple, relatable examples.
4. Involve the students through short questions on implicit differentiation.
5. Solve a numerical on higher order derivatives together with the class.
6. Summarize the lesson and highlight the key points of Differentiation.

Classroom Activities

• Group discussion: students compare rules of differentiation with chain rule.

• Board activity: solve a question on implicit differentiation in pairs.

• Quick quiz on the key terms of Differentiation.

Assessment Questions

1. Define derivative definition.
2. Differentiate between rules of differentiation and chain rule.
3. Solve a numerical based on implicit differentiation.
4. Write a short note on higher order derivatives.

Homework

• Read Differentiation from the book and solve the exercise questions.

• Prepare short notes on derivative definition and rules of differentiation.

Time Allocation

4–6 periods of 40 minutes each

Unit 3: Integration

Learning Objectives

• Define and explain indefinite integrals in the context of Integration.

• Differentiate between definite integrals and substitution method with suitable examples.

• Solve numerical problems related to integration by parts accurately.

• Apply the ideas of Integration to everyday situations involving area under curve.

Key Concepts

• Meaning, definition and scope of indefinite integrals.

• Important properties, types and examples of definite integrals.

• Relationship between substitution method and integration by parts.

• Practical applications of Integration, especially area under curve.

• Key terms and a summary of the main points of Integration.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Integration with a daily-life example.
2. Explain indefinite integrals on the board with definitions and clear diagrams.
3. Discuss definite integrals and substitution method using simple, relatable examples.
4. Involve the students through short questions on integration by parts.
5. Solve a numerical on area under curve together with the class.
6. Summarize the lesson and highlight the key points of Integration.

Classroom Activities

• Group discussion: students compare definite integrals with substitution method.

• Board activity: solve a question on integration by parts in pairs.

• Quick quiz on the key terms of Integration.

Assessment Questions

1. Define indefinite integrals.
2. Differentiate between definite integrals and substitution method.
3. Solve a numerical based on integration by parts.
4. Write a short note on area under curve.

Homework

• Read Integration from the book and solve the exercise questions.

• Prepare short notes on indefinite integrals and definite integrals.

Time Allocation

4–6 periods of 40 minutes each

Unit 4: Introduction to Analytic Geometry

Learning Objectives

• Define and explain coordinate system in the context of Introduction to Analytic Geometry.

• Differentiate between distance formula and slope with suitable examples.

• Solve numerical problems related to equation of line accurately.

• Apply the ideas of Introduction to Analytic Geometry to everyday situations involving parallel lines.

Key Concepts

• Meaning, definition and scope of coordinate system.

• Important properties, types and examples of distance formula.

• Relationship between slope and equation of line.

• Practical applications of Introduction to Analytic Geometry, especially parallel lines.

• Key terms and a summary of the main points of Introduction to Analytic Geometry.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Introduction to Analytic Geometry with a daily-life example.

2. Explain coordinate system on the board with definitions and clear diagrams.

3. Discuss distance formula and slope using simple, relatable examples.

4. Involve the students through short questions on equation of line.

5. Solve a numerical on parallel lines together with the class.

6. Summarize the lesson and highlight the key points of Introduction to Analytic Geometry.

Classroom Activities

• Group discussion: students compare distance formula with slope.

• Board activity: solve a question on equation of line in pairs.

• Quick quiz on the key terms of Introduction to Analytic Geometry.

Assessment Questions

1. Define coordinate system.

2. Differentiate between distance formula and slope.

3. Solve a numerical based on equation of line.

4. Write a short note on parallel lines.

Homework

• Read Introduction to Analytic Geometry from the book and solve the exercise questions.

• Prepare short notes on coordinate system and distance formula.

Time Allocation

4–6 periods of 40 minutes each

Unit 5: Linear Inequalities and Linear Programming

Learning Objectives

• Define and explain inequalities in the context of Linear Inequalities and Linear Programming.

• Differentiate between feasible region and objective function with suitable examples.

• Solve numerical problems related to graphical solution accurately.

• Apply the ideas of Linear Inequalities and Linear Programming to everyday situations involving optimization.

Key Concepts

• Meaning, definition and scope of inequalities.

• Important properties, types and examples of feasible region.

• Relationship between objective function and graphical solution.

• Practical applications of Linear Inequalities and Linear Programming, especially optimization.

• Key terms and a summary of the main points of Linear Inequalities and Linear Programming.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Linear Inequalities and Linear Programming with a daily-life example.
2. Explain inequalities on the board with definitions and clear diagrams.
3. Discuss feasible region and objective function using simple, relatable examples.
4. Involve the students through short questions on graphical solution.
5. Solve a numerical on optimization together with the class.
6. Summarize the lesson and highlight the key points of Linear Inequalities and Linear Programming.

Classroom Activities

• Group discussion: students compare feasible region with objective function.

• Board activity: solve a question on graphical solution in pairs.

• Quick quiz on the key terms of Linear Inequalities and Linear Programming.

Assessment Questions

1. Define inequalities.
2. Differentiate between feasible region and objective function.
3. Solve a numerical based on graphical solution.
4. Write a short note on optimization.

Homework

• Read Linear Inequalities and Linear Programming from the book and solve the exercise questions.

• Prepare short notes on inequalities and feasible region.

Time Allocation

4–6 periods of 40 minutes each

Unit 6: Conic Sections

Learning Objectives

• Define and explain circle in the context of Conic Sections.

• Differentiate between parabola and ellipse with suitable examples.

• Solve numerical problems related to hyperbola accurately.

• Apply the ideas of Conic Sections to everyday situations involving standard equations.

Key Concepts

• Meaning, definition and scope of circle.

• Important properties, types and examples of parabola.

• Relationship between ellipse and hyperbola.

• Practical applications of Conic Sections, especially standard equations.

• Key terms and a summary of the main points of Conic Sections.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Conic Sections with a daily-life example.
2. Explain circle on the board with definitions and clear diagrams.
3. Discuss parabola and ellipse using simple, relatable examples.
4. Involve the students through short questions on hyperbola.
5. Solve a numerical on standard equations together with the class.
6. Summarize the lesson and highlight the key points of Conic Sections.

Classroom Activities

• Group discussion: students compare parabola with ellipse.

• Board activity: solve a question on hyperbola in pairs.

• Quick quiz on the key terms of Conic Sections.

Assessment Questions

1. Define circle.
2. Differentiate between parabola and ellipse.
3. Solve a numerical based on hyperbola.
4. Write a short note on standard equations.

Homework

• Read Conic Sections from the book and solve the exercise questions.

• Prepare short notes on circle and parabola.

Time Allocation

4–6 periods of 40 minutes each

Unit 7: Vectors

Learning Objectives

• Define and explain vector in space in the context of Vectors.

• Differentiate between dot product and cross product with suitable examples.

• Solve numerical problems related to triple products accurately.

• Apply the ideas of Vectors to everyday situations involving applications.

Key Concepts

• Meaning, definition and scope of vector in space.

• Important properties, types and examples of dot product.

• Relationship between cross product and triple products.

• Practical applications of Vectors, especially applications.

• Key terms and a summary of the main points of Vectors.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Vectors with a daily-life example.
2. Explain vector in space on the board with definitions and clear diagrams.
3. Discuss dot product and cross product using simple, relatable examples.
4. Involve the students through short questions on triple products.
5. Solve a numerical on applications together with the class.
6. Summarize the lesson and highlight the key points of Vectors.

Classroom Activities

• Group discussion: students compare dot product with cross product.

• Board activity: solve a question on triple products in pairs.

• Quick quiz on the key terms of Vectors.

Assessment Questions

1. Define vector in space.
2. Differentiate between dot product and cross product.
3. Solve a numerical based on triple products.
4. Write a short note on applications.

Homework

• Read Vectors from the book and solve the exercise questions.

• Prepare short notes on vector in space and dot product.

Time Allocation

4–6 periods of 40 minutes each

Unit 8: Applications of Differentiation and Integration

Learning Objectives

• Define and explain maxima and minima in the context of Applications of Differentiation and Integration.

• Differentiate between rate of change and tangent and normal with suitable examples.

• Solve numerical problems related to area problems accurately.

• Apply the ideas of Applications of Differentiation and Integration to everyday situations involving volume.

Key Concepts

• Meaning, definition and scope of maxima and minima.

• Important properties, types and examples of rate of change.

• Relationship between tangent and normal and area problems.

• Practical applications of Applications of Differentiation and Integration, especially volume.

• Key terms and a summary of the main points of Applications of Differentiation and Integration.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Applications of Differentiation and Integration with a daily-life example.

2. Explain maxima and minima on the board with definitions and clear diagrams.

3. Discuss rate of change and tangent and normal using simple, relatable examples.

4. Involve the students through short questions on area problems.

5. Solve a numerical on volume together with the class.

6. Summarize the lesson and highlight the key points of Applications of Differentiation and Integration.

Classroom Activities

• Group discussion: students compare rate of change with tangent and normal.

• Board activity: solve a question on area problems in pairs.

• Quick quiz on the key terms of Applications of Differentiation and Integration.

Assessment Questions

1. Define maxima and minima.

2. Differentiate between rate of change and tangent and normal.

3. Solve a numerical based on area problems.

4. Write a short note on volume.

Homework

• Read Applications of Differentiation and Integration from the book and solve the exercise questions.

• Prepare short notes on maxima and minima and rate of change.

Time Allocation

4–6 periods of 40 minutes each